

Figure Catalog

CytoCV Research Documentation

Purpose

This catalog summarizes the current CytoCV diagram set for reports, appendices, collaborator packets, and research-facing documentation. It uses the same public terminology as the current interface: DIC, Blue, Red, and Green.

Primary Figures

Figure	Recommended use	Caption focus
Figure 1. System architecture	Overall software architecture	Layered view of the web interface, scientific processing components, and persistent storage boundaries.
Figure 2. End-to-end workflow	Methods overview	DeltaVision ingestion through validation, segmentation, quantification, review, and export.
Figure 3. Cell analysis flow	Per-cell measurement section	How DIC-based segmentation and channel-specific measurements combine at the single-cell level.
Figure 4. Data model	Reproducibility or implementation appendix	Relationships among uploaded runs, previews, segmented outputs, and per-cell statistics.

Supplementary Figures

Topic group	Figures	Typical use
Validation and channel rules	Plugin-channel map; upload validation flow; scale and channel resolution	Appendix material for required-channel logic, metadata interpretation, and scale handling.
Processing and output	Preprocess and inference flow; segmentation output flow; display and export flow	Detailed workflow explanation beyond the main narrative figure set.
Retention and control	Artifact lifecycle; progress and cancellation state; run ownership and retention state	Operational context for run persistence, cancellation, and saved-result handling.
Access and legacy context	Authentication and account flow; legacy Blue measurements	Account-based access overview and legacy analysis behavior when Blue-channel workflows are still needed.

Figure Usage Notes

- Use the first four figures for the primary narrative in a manuscript, thesis chapter, or software appendix.
- Use the supplementary figures when reviewers or collaborators need more detail on validation rules, workflow control, or legacy analysis paths.
- Keep figure captions aligned with the current public channel terms Blue, Red, Green, and DIC rather than older instrument-specific names.
- Revise the figure set whenever the workflow, validation policy, or measurement terminology changes.

Maintenance Note

Editable diagram sources and rendered PNG assets are maintained alongside the rest of the repository documentation so that captions and diagrams can be updated together.